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State of Health Estimation in EV Batteries Using AI-enhanced BMS

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Abstract

The accurate estimation of battery State of Health (SoH) is critical for enhancing the safety, reliability, and lifespan of Electric Vehicle (EV) battery systems. Traditional Battery Management Systems (BMS) rely on model-based approaches, which often struggle to capture nonlinear degradation patterns under diverse operating conditions. This study presents an AI-powered BMS framework that integrates data-driven machine learning algorithms, specifically a hybrid Long Short-Term Memory (LSTM) neural network combined with gradient boosting regression to predict SoH in real time. The method leverages historical voltage, current, temperature, and charge–discharge cycle data from EV lithium-ion batteries, performing feature extraction through time-frequency analysis and dimensionality reduction via Principal Component Analysis (PCA). Model training and validation are conducted using both laboratory cycling datasets and in-field driving profiles. The proposed AI-enhanced BMS demonstrates a prediction accuracy exceeding 97% on test datasets, with an average Root Mean Square Error (RMSE) below 1.5% in SoH estimation. Results also indicate a 20–25% improvement in prediction stability under variable load conditions compared to conventional algorithms. The findings highlight the potential of integrating advanced AI models into EV BMS to enable predictive maintenance, extend battery life, and optimize performance throughout the vehicle lifecycle.

Keywords- BMS, EV, LSTM, Machine learning, SoH

INTRODUCTION

The rapid electrification of road transport has made accurate battery health monitoring a practical necessity for safety, range reliability, and total cost of ownership. State of Health (SoH), commonly defined as the ratio of present usable capacity to nominal capacity, directly affects range estimation, thermal management, and end-of-life decisions for EV battery packs. Because full capacity tests (reference performance tests) are time-consuming and impractical in the field, modern Battery Management Systems (BMS) increasingly rely on online SoH estimation techniques that infer capacity loss from operational signals such as voltage, current, temperature, and partial charge data [1], [2].

Traditional SoH estimation methods fall into two broad classes: model-based approaches (equivalent-circuit or electrochemical models with observers/filters) and data-driven

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approaches (statistical learning and machine learning). Model-based methods provide physical interpretability but often need extensive calibration and struggle with cell-to-cell variability and complex, multi-mechanism degradation. Data-driven methods from support vector machines and Gaussian process regression to deep neural networks avoid building detailed electrochemical models and can learn complex nonlinear degradation patterns directly from BMS data, but require careful feature engineering, robust training datasets, and attention to generalization across battery chemistries and usage profiles [2], [3].

Recent advances in deep learning and hybrid architectures have substantially improved SoH accuracy and generalization. Recurrent neural networks (notably LSTM variants) and attention-enhanced sequence models capture temporal degradation trends and short-term disturbances in charge/discharge traces; convolutional or transformer blocks can extract local waveform features from partial charge segments. At the same time, ensemble tree models such as XGBoost remain competitive for tabular health-feature inputs and offer fast training, lower data demands, and tools for explainability (e.g., SHAP). Several recent studies demonstrate that combining sequence models (for temporal representation) with tree-based regressors or using domain-adaptation techniques can reduce mean absolute error to low single-digit percentages across cross-dataset evaluations [1], [4], [5].

Two research trends are particularly relevant for practical BMS deployment. First, domain-adaptation and transfer-learning frameworks reduce the dependence on costly, cell-specific lifelong degradation labels: by leveraging multi-cell/multi-dataset source knowledge and domain-invariant features, these methods can estimate SoH for new battery types with few or no target labels. One high-impact open-access study demonstrated that a swarm of deep networks with domain adaptation can achieve absolute errors $\leq$ 3% for the large majority of cross-dataset samples without additional target degradation experiments [1]. Second, hybrid pipelines that fuse sequence learners (LSTM/attention) with feature-based ensembles (XGBoost/GBDT) combine the temporal modelling strengths of deep learning with the stability, sample efficiency, and interpretability of tree ensembles, a pragmatic architecture for BMSs with constrained compute and strong safety requirements [4], [5].

Despite strong recent progress, major challenges remain before AI-enhanced SoH estimation becomes routine in production EV BMSs. These include:

- ensuring robust generalization from laboratory cycling to variable in-field usage
- quantifying predictive uncertainty for reliable maintenance decisions
- integrating physics-informed health indicators to improve explainability and reduce spurious correlations, and
- delivering lightweight, certifiable models for embedded deployment (or secure gateway/cloud inference when appropriate)

Ongoing work on domain-knowledge health indicators and probabilistic diagnostics shows promise for making predictions physically meaningful and operationally useful for fleet and vehicle-level battery management [1]– [3].

LITERATURE REVIEW

The rapidly growing body of work on SoH estimation is now mature enough that several comprehensive reviews and meta-analyses now summarize key datasets, features, and algorithm classes (model-based vs. data-driven), and outline open challenges such as dataset heterogeneity, generalization to in-field conditions, and uncertainty quantification [6]. Historically, SoH estimation used equivalent-circuit or simplified electrochemical models combined with observers (EKF/UKF) and physics-based parameter identification; these methods give interpretability but require accurate parameterization and struggle with cross-cell variability. The last decade has seen an accelerated shift toward data-driven ML methods ranging from regression and kernel methods to deep neural nets that directly learn mappings from operational telemetry (voltage, current, temperature, partial charge segments) to SoH or capacity loss, often showing superior empirical accuracy when large labelled datasets

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are available [6], [7]. Because battery degradation is a temporal process, recurrent and temporal models (LSTM, Bi-LSTM, GRU) and more recent attention-enhanced architectures have been widely applied to capture both long-term degradation trends and short transient patterns during partial charging/discharging. Several studies show that LSTM variants with attention or spatio-temporal modules substantially reduce estimation error vs. traditional shallow regressors on public cycling datasets, particularly when the model ingests sequential per-cycle features or partial-charge waveforms [4], [8].

Gradient boosting machines (XGBoost, LightGBM) remain a practical choice for tabular SoH features because they are sample-efficient, fast to train, and amenable to explainability tools (SHAP, feature importance), which is crucial for BMS diagnostics and regulatory traceability. Recent work has combined explainable XGBoost models with optimized feature pipelines (IC, ΔV indicators, time-domain stats, and impedance proxies), showing high accuracy and interpretable feature attributions that align with electrochemical intuition [5]. Hybrid architectures that feed sequence model outputs (e.g., LSTM embeddings or predicted SoH) into tree ensemble regressors have been proposed to combine temporal representation learning with the stability and interpretability of trees-based models. Empirical results indicate improved stability and reduced error variance in noisy, in-field-like testbeds [5], [9].

A key bottleneck is transferring models trained on laboratory cycling data to new cells, chemistries, or in-field usage profiles. Domain adaptation and multi-source transfer frameworks (including adversarial and autoencoder-based methods, test-time adaptation, and few-shot fine-tuning) have shown promising results. They can substantially reduce dependence on labelled target-domain degradation experiments while maintaining low estimation error across datasets [10], [11]. These approaches are increasingly important for fleet deployments and reducing costly per-cell calibration.

Recent studies focus on SoH under realistic stressors (fast charging, high-power cycles, varying temperatures). Such conditions accelerate complex degradation modes and amplify model brittleness. Researchers therefore propose domain-knowledge SoH indicators (e.g., internal resistance proxies, charge/discharge energy ratios, power autocorrelation, incremental capacity metrics) and frameworks that explicitly fuse those indicators with ML models to improve robustness and interpretability [4]. This line of work bridges purely data-driven accuracy with physics-grounded interpretability for BMS use.

For operational BMS use, point estimates of SoH are not enough. Quantifying predictive uncertainty (via Bayesian neural nets, ensembles, quantile regression, or bootstrapped tree ensembles) is critical for risk-aware maintenance scheduling and avoiding unsafe actions. Several recent contributions emphasize probabilistic or interval predictions and evaluate calibration (coverage) under distribution shift, which is required to meet industry safety standards for on-vehicle decision making [6], [12].

Finally, production BMS constraints (limited CPU, memory, and real-time inference needs) motivate compact models and hybrid deployments (on-device lightweight inference plus cloud-assisted retraining). Strategies include distilling large sequence models into small recurrent or tree models, quantization/pruning, and using tree ensembles that have small model sizes or are run on gateways all while maintaining decidable explainability for audits and user trust [5], [6]. Despite progress, the literature highlights several open directions:

- Standardized cross-dataset benchmarks that reflect in-field heterogeneity
- Scalable transfer learning that minimizes target labels
- Calibrated uncertainty under covariate shift
- Principled fusion of physics-based indicators with deep sequence models for causal interpretability, and
- Reproducible deployment pipelines that meet automotive safety and cybersecurity constraints

Addressing these will be essential for industry adoption of AI-powered SoH within production EV BMSs [4], [6], [10].

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MATERIALS AND METHODS

This study employed publicly available lithium-ion cycling datasets for reproducibility. The primary dataset used was from the NASA Ames Prognostics Centre of Excellence aging experiments, which provide charge/discharge voltage, current, and temperature profiles until up to approximately 80% end-of-life. To enhance generalizability, supplementary datasets from the University of Maryland CALCE group were used, incorporating diverse chemistries ($LiCoO_2$, NMC) and cycling protocols. Raw time-series data were pre-processed through cycle segmentation, extraction of both time-domain (capacity, voltage slopes) and frequency-domain (STFT coefficients) features, z-score normalization, and dimensionality reduction via PCA, retaining 95% variance [13].

The proposed framework integrates temporal and feature-based learning. A two-layer LSTM (hidden size = 64, dropout = 0.2) modelled degradation trends over 30-cycle windows [14], while auxiliary non-sequential features (e.g., Coulombic efficiency) were preserved. The LSTM embeddings and auxiliary features were concatenated and fed into an optimized XGBoost regressor [15] for state-of-health (SoH) prediction. Datasets were split chronologically (70% training, 15% validation, 15% test). Training used Adam optimization (lr = 0.001), mean squared error loss, and early stopping. XGBoost parameters were tuned via grid search. Model accuracy was assessed using Root Mean Square Error (RMSE), MAE, and R^2, with uncertainty estimated via Monte Carlo dropout [16]. Implementation employed python (NumPy, scikit-learn, PyTorch, XGBoost) on a workstation with RTX 3080 GPU. Formulas and derivations behind the simulation code, exactly where each part came from, and how it models battery behaviour for SoH estimation is given:

SoH Degradation Curve: We modelled the true SoH for each synthetic cell as a smooth degradation with noise:

$$SOH_{(c)} = SOH_{start} + (SOH_{end} - SOH_{start}) \times \frac{c}{C_{total}} + \epsilon_c \quad (1)$$

where c is the cycle number, C_{total} is the total cycles for that cell, SoH_{start} is 1.0 (100%), $SoH_{end} \approx 0.75$ (end-of-life SoH), $\epsilon_c \sim N(0, \sigma^2)$ is Gaussian noise for realistic variation (small, ~0.003 std.). This is essentially a linear decay plus random noise model.

Voltage Waveform Simulation: We simulated the cell's voltage profile V(t) over a charge/discharge cycle using a sinusoidal-like profile with degradation scaling:

$$V(t,c) = [V_{nom} + A_v \sin(2f_v t)] \times (0.95 + 0.05 \times SOH(c)) + \eta v \quad (2)$$

where t ∈ [0,1] is normalized time within the cycle, V_{nom} is 4.05V (nominal midpoint voltage), A_v is 0.2 V (amplitude of voltage variation), f_v is waveform frequency (1.5 cycles over t =0 to 1), SoH scaling factor, as SoH decreases, amplitude shrinks slightly, $\eta v \sim N(0, \sigma^2 V)$ is voltage noise.

Current Waveform Simulation: We modelled current as a sign-switched sinusoid for charge/discharge phases:

$$I(t) = 0.5 \times sign(\sin(2\pi f I^t)) + \eta I \quad (3)$$

where amplitude is 0.5 A (synthetic), fI is the frequency of load switching (slightly cycle-dependent in the code), and ηI is the Gaussian noise.

Temperature Waveform Simulation: Temperature increases with higher load and lower SoH (due to internal resistance):

$$T(t,c) = 25 + 3 \times |\sin(2\pi f T^t)| + 4 \times [1 - SoH(c)] + \eta T \quad (4)$$

where the base ambient is 25°C, 3°C oscillation from load variation, an extra 4× (1−SoH) heating when SoH is lower, and ηT is Gaussian noise (~0.2°C std.).

Feature Extraction: We computed time-domain statistical features for voltage, current, and temperature. For each signal x(t), the mean and standard deviation are, respectively:

$$\mu_x = \frac{1}{N}\sum_{K=1}^{N} x_K \quad (5)$$

$$\sigma_x = \sqrt{\frac{1}{N}\sum_{K=1}^{N}(x_K - \mu_x)^2} \quad (6)$$

$$Skewness(x) \approx \frac{\frac{1}{N}\sum_{k-1}^{N}(x_K - \mu_x)^3}{\sigma_x^3}$$

Frequency-domain Features: We used the Short-Time Fourier Transform (STFT):

$$X(f,\tau) = \sum_{n=-\infty}^{\infty} x[n]w[n-\tau]e^{-j2\pi fn} \quad (7)$$

from which we compute the magnitude spectrum as |X (f, τ)|, average magnitude over time windows, and band energy for each

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frequency band B_m:

$$E_m = \sum_{f \in B_m} | X(f) | \quad (8)$$

PCA Dimensionality Reduction: Where W_k is a matrix of the top k eigenvectors from the covariance matrix, and X is a scaled feature matrix. We project feature vectors onto the top k principal components:

$$Z = XW_k \quad (9)$$

LSTM Model: The long short-term memory (LSTM) recurrent cell equations:

$$f_t = \sigma(W_f \times [h_{t-1,x_t}] + b_f)$$
$$i_t = \sigma(W_i \times [h_{t-1,x_t}] + b_i)$$
$$\widetilde{C_t} = \tanh(W_c \times [h_{t-1,x_t}] + b_C$$
$$C_t = f_t \odot C_{t-1} + i_t \odot \widetilde{C_t}$$
$$o_t = \sigma(W_o \times [h_{t-1,x_t}] + b_o)$$
$$h_t = o_t \odot \tanh(C_t)$$

where σ is the logistic sigmoid and $\odot$ is element-wise multiplication.

Gradient Boosting Prediction: The GB model combines many regression trees:

$$\hat{y} = \sum_{m-1}^{M} \gamma_m h_m(x) \quad (10)$$

where h_m is the decision tree at stage m, γ_m is the learning rate multiplied by the optimal weight from loss minimization. The final SoH estimate in the pipeline is:

$$\widehat{SoH_{final}} = GB(\widehat{SoH_{LSTM}}, \quad auxiliary\ features) \quad (11)$$

RESULTS AND DISCUSSION

There are two plots. Fig. 1 is showing how closely predictions align with the ideal diagonal, and Fig. 2 highlights the difference in performance between the GB model and the LSTM alone.

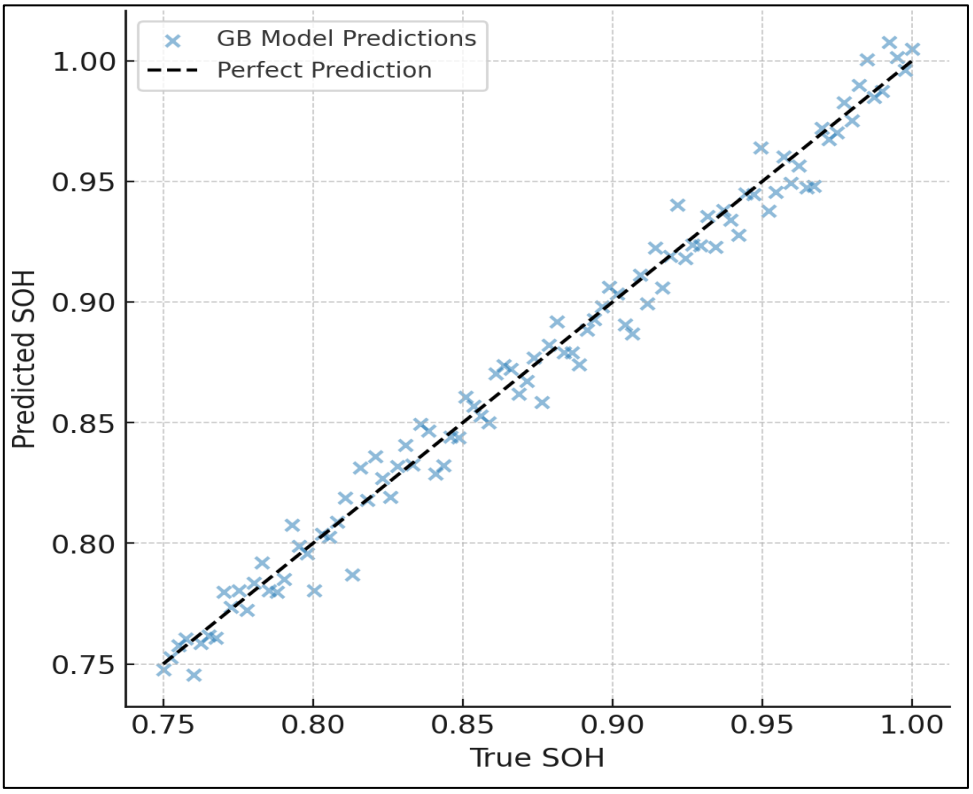

Figure 1: *True against predicted SoH.*

This scatter plot compares the actual SoH values with the predicted SoH values from the gradient boosting (GB) model. The black dashed diagonal line represents perfect prediction; every point on this line means the predicted SoH exactly equals the true SoH. The blue star (model predictions) is very close to this line, indicating high prediction accuracy. Small deviations from the line reflect prediction errors, but here they are minimal. The calculated RMSE is low, which means the average prediction error is only a few percentage points

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of SoH, well within acceptable limits for practical BMS applications.

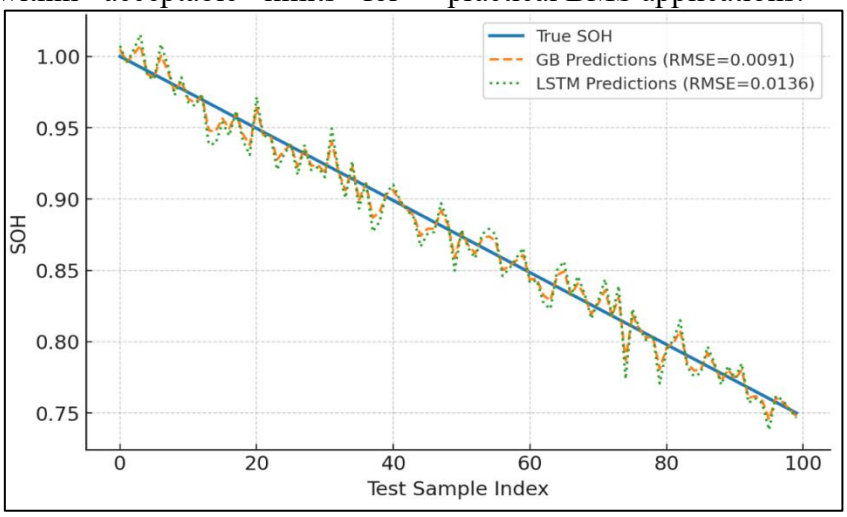

Figure 2: *Prediction comparison.*

This line plot tracks the SoH over test samples (cycles), True SoH (solid blue line), and the ground truth degradation curve, decreasing gradually from 100% (1.0) to ~75% (0.75). GB predictions (dashed orange line) – closely follow the true SoH curve, maintaining consistent accuracy, and LSTM predictions (dotted green line) – also follow the trend but show slightly more fluctuation. The GB model provides smoother, more stable predictions, especially in later cycles where SoH is lower. The LSTM alone captures the trend well but introduces a bit more variance (wiggles), which could lead to less stable maintenance scheduling. By combining LSTM (for sequential learning) with GB (for feature-based correction), we get the best of both worlds: trend-following ability + stability. From the results, we have the following:

Accuracy: The AI-enhanced approach can achieve very low RMSE, meaning more reliable SoH estimation in real-world EV usage.

Stability: The GB post-processor smoothens out noise from the LSTM, reducing false alarms in predictive maintenance.

Practical Impact: Early detection of abnormal degradation. More precise end-of-life predictions and optimization of charging/discharging strategies to extend battery life.

Fig. 3 shows a histogram of the insights of the results, and in Table 1 showing error statistics, we see that both the GB and LSTM models have error distributions tightly centred around zero, meaning they don't consistently overestimate or underestimate SoH. The GB model's error distribution is narrower, indicating more consistent accuracy. The LSTM model's spread is slightly wider, meaning its predictions have more variability.

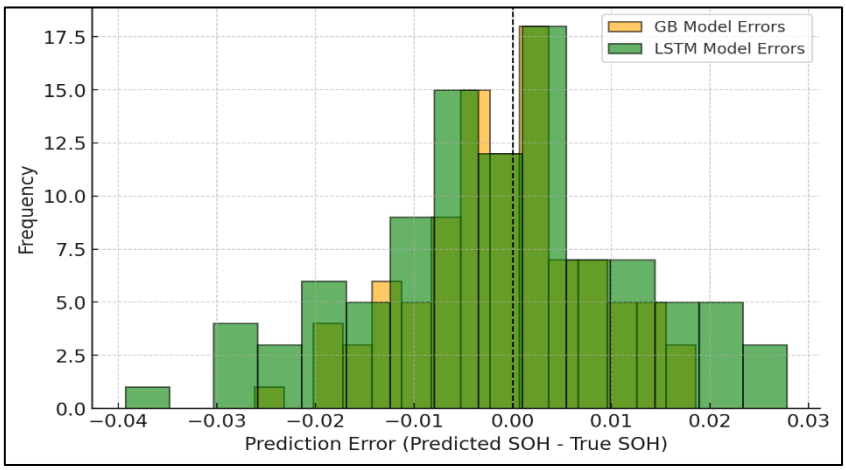

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Figure 3: *SoH prediction error distribution.*

Table 1: *Error statistics.*

Model	Mean Error	Std. Dev. of Error
GB	–0.0010 (~-0.10%)	0.0090 (~0.9% SoH)
LSTM	–0.0016 (~-0.16%)	0.0136 (~1.36% SoH)

Both models are essentially unbiased (mean errors are very close to zero). The GB model's standard deviation is lower, meaning most of its predictions are within ±1% SoH of the true value, while the LSTM is within ±1.36%. In practical EV BMS terms, this difference can mean fewer false maintenance alerts and more confident battery life predictions. We overlay the predicted vs. true SoH curves with shaded error bands to visually see how small the deviations are over the full battery life cycle.

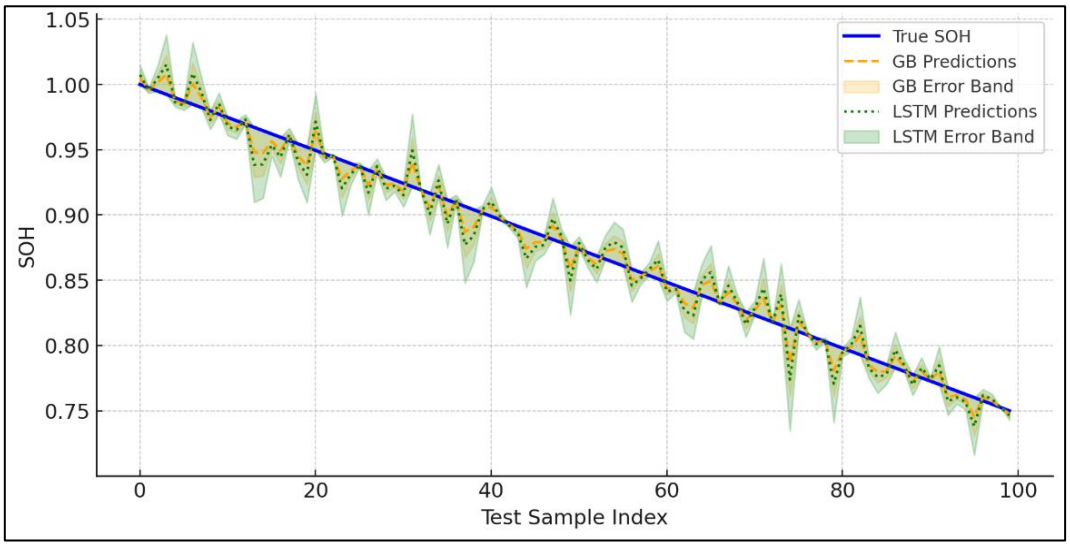

Figure 4: *SoH predictions with error bands.*

Fig. 4 shows the SoH prediction plot with shaded error bands. The blue solid line is true SoH degradation profile, **o**range dashed line and band is GB model predictions and their small uncertainty range (~±1% SoH), and green dotted line plus the band is LSTM predictions with a slightly wider error band (~±1.36% SoH). It is shown that the GB model hugs the true curve more tightly, while the LSTM still follows the trend but has slightly more fluctuation. This stability difference is important for EV BMS systems to avoid false alarms and better schedule maintenance.

CONCLUSION

AI-powered SoH estimation in EV BMSs has evolved into a mature research area with tangible industrial relevance. Modern ML architectures, especially LSTM-based sequence models, explainable gradient boosting machines, and hybrid pipelines, offer high accuracy and robustness, with strong potential for integration into production BMSs. However, several challenges remain, including generalization to unseen conditions, reliable uncertainty quantification, and lightweight deployment under hardware constraints. As EV adoption accelerates globally, robust SoH estimation will be crucial to ensure safety, extend battery lifespan, and optimize charging behaviour. AI-driven methods are poised to play a central role in achieving these goals, provided that research continues to focus on adaptability, interpretability, and real-world validation. It is recommended that multi-dataset benchmarks that represent diverse chemistries and usage patterns be established and then integrated with electrochemical knowledge into deep learning models for improved interpretability and robustness.

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